

BIP 360 Conformance Lab — Phase 1

Independent, version-pinned conformance testing and cross-implementation differential analysis for BIP 360 (Pay-to-Merkle-Root), ending in the first independent implementation-conformance report on a quantum-resistant Bitcoin transaction proposal

Program Galaxy Bitcoin Quantum Readiness Initiative

Priority area #1 — implementation and review of quantum-resistant transaction proposals **Ask** USD 45,000 · 12 weeks · 4 gates

First tranche USD 5,000 (retroactive, for the already-public baseline) **Applicant** Ashwin Goyal · individual · Bhopal, India

License MIT · fully open source

This revises my earlier application ("P2MR Assurance Lab", USD 400,000 / 28 weeks). On reflection, a first engagement between a new grant program and a first-time grantee should be small, fast, and verifiable. This is that engagement; the larger scope survives as an optional Phase 2 (\$7) that I will propose only after — and only if — Phase 1 is accepted as delivered.

0 Verify before reading

```
# Public baseline — clone and run (Python 3.10+, no dependencies):
git clone https://github.com/let-the-dreamers-rise/p2mr-assurance-lab
cd p2mr-assurance-lab && python verify_vectors.py && python run_mutations.py

# Expected:
9/9 official BIP 360 construction vectors PASS (byte-for-byte)
8/8 local boundary tests PASS
24/24 adversarial mutation cases PASS
pinned: bitcoin/bips @ ed4ffcb6a48d4dc4fdcf11cdba783c233db8c66e
fixture: p2mr_construction.json · sha256: cd02da0b3c5bcbea98ed4d3141ebaf71344ee5acc8eb543841b7d2ed8532
51c7
```

If the repository is missing, the pin is absent, or any vector fails on your machine — reject this application. Nothing below asks you to believe anything that command does not already demonstrate.

1 What USD 45,000 buys

DELIVERABLE

WHAT IT IS, CONCRETELY

1. Pinned adversarial conformance corpus (250+ cases)	Mutations across every P2MR construction surface — BIP 340 tagged hashing, tree shape, CompactSize boundaries, control-block ordering and parsing, leaf-version masking, Bech32m, depth/length limits — each case carrying its expected result, a BIP-text citation, and a minimized reproducer. Reusable by every present and future implementer; pinned so results never rot against the moving Draft.
2. Cross-implementation differential matrix	The corpus run against the bips reference implementation, the open bitcoinjs-lib P2MR branch, and available Bitcoin Core regtest patches, each pinned to exact upstream commits. States are strictly pass / fail / not-implemented / not-buildable / not-applicable. Every security-relevant divergence goes to maintainers through coordinated disclosure before publication.
3. Property-based and fuzz harness, in CI	Continuous randomized construction/parse round-trip testing with recorded seeds; CI refuses unpinned runs and records BIP commit, fixture hash, platform, and counts on every run.
4. Offline exposure-classifier CLI (v1)	A local-only, no-telemetry tool classifying output scripts by quantum-exposure class (\$6 taxonomy) from non-secret exports; hard-refuses seeds, xprv, WIF, and wallet databases — refusal behavior is itself tested. The seed of the custodian migration tooling in Phase 2.
5. "State of BIP 360 Implementation Conformance" — the report	A publication-grade, fully reproducible report: methodology, findings, divergence analysis, disclosure outcomes, and open questions for the draft's evolution. Galaxy receives pre-publication review and, at its option, first publication through Galaxy Research — the first independent conformance assessment of any quantum-resistant Bitcoin transaction proposal.

2 Why this is a grant worth making — from your side of the table

- **It is your priority area #1, already started.** The baseline exists, is public, and is reproducible in ten minutes. You are not funding an intention; you are extending a running project.
- **It fills a gap no one else occupies.** Implementers test their own code; nobody funds independent adversarial conformance work on a draft. Yet the draft stage is exactly when byte-level divergence is cheap to fix — and P2MR's machinery (control blocks, CompactSize, leaf versions, Bech32m) is exactly where a one-byte mismatch strands funds after activation.
- **Independence is the point.** Differential testing is most credible performed outside every implementing team. A funded independent tester is a structural asset the implementations cannot provide for themselves.
- **A new program needs early, legible wins.** This grant is small, fast, fully open source, zero custody/regulatory surface, and ends in a citable report your research arm may publish first. It is the shape of grant a three-week-old program can point to in its first progress update.
- **Administration is bounded.** Four gates, each reviewable in under an hour against a one-page acceptance sheet mapping every criterion to a command, a file, or a diff.
- **Diligence shortcut.** Forward this PDF to the BIP 360 author or any implementation maintainer and ask one question: "would an independent, version-pinned conformance corpus and differential test matrix be useful right now?" Their answer is a better signal than anything I can write here.

3 Milestones and payment

GATE	WEEK	PAYMENT	CUMULATIVE	ACCEPTED AGAINST
M0	—	\$5,000	\$5,000	The already-public baseline: pinned verifier (9/9 official vectors, 8/8 boundary tests), scope charter, public funds ledger. Retroactive — paid only if you accept delivered work.
M1	3	\$10,000	\$15,000	100+ case mutation corpus with spec citations and minimized reproducers; threat model; coordinated-disclosure policy; clean-clone reproducibility report.
M2	8	\$15,000	\$30,000	250+ case corpus; differential matrix across all three pinned implementation targets; fuzz/property harness in CI; any security-relevant divergence disclosed to maintainers.
M3	12	\$15,000	\$45,000	Exposure-classifier CLI v1 with refusal tests; the final conformance report, reproducible end-to-end, with Galaxy pre-publication review; six months of pin/corpus maintenance committed at no further cost.

Budget: \$36,000 lead engineering (12 weeks, full-time, sole applicant) · \$6,000 CI, fuzzing, and build infrastructure · \$3,000 disclosure and release operations. No other categories. **Stop rights:** you may accept, reject, pause, or terminate at any gate; nothing advances automatically; a blocked milestone is reported as blocked, never relabeled complete. Grant funds never touch trading, tokens, custody, marketing, or undisclosed subcontractors.

4 Who is asking

Ashwin Goyal — B.Tech computer science undergraduate, VIT Bhopal, India; open-source engineer, ~90 public repositories (prior shipped work includes a deterministic transaction-policy engine with signed receipts and a live milestone-gated oracle demo — shown as evidence of shipping discipline, not Bitcoin credentials). No Bitcoin Core commits, no published cryptography research, one prior \$200 developer microgrant. Weight my credentials at zero: the baseline artifact, the pins, the gates, and your stop rights carry this application — you never pay ahead of work you have already accepted.

5 Boundaries and claim discipline

This banner ships verbatim on every funded artifact:

"BIP 360 (P2MR) is a Draft and is not activated on Bitcoin mainnet. P2MR can mitigate long-exposure (revealed-key) risk only under fresh-key discipline; it does not by itself defend the reveal-to-confirmation window, and nothing in this work makes Bitcoin quantum-safe today."

- No consensus change, activation claim, or representation as a Bitcoin Core release; no key material, custody, signing, broadcasting, telemetry, or mainnet flows; no token, bridge, or trading product.
- "Audit" is reserved for contracted qualified external auditors — never for my own testing.
- Coordinated disclosure: pin and reproduce, minimize, notify maintainers privately, allow a remediation window, publish facts only, retest — a compatibility bug is never converted into a claim that Bitcoin is broken.
- Any public release that names Galaxy is submitted to Galaxy for review before publication.
- **Draft risk:** if BIP 360 is materially revised or superseded mid-program, the pinned corpus and version-diff tracker retarget to the successor revision — the methodology is proposal-portable — or you stop at the next gate. That is what the gates are for.

6 Exposure taxonomy used by the classifier

CLASS	EXAMPLES	PERMITTED OUTPUT
Direct EC public-key exposure	P2TR, bare P2PK, bare multisig	Prioritized for long-exposure analysis; the tool never states funds are compromised.
Reuse/history required	P2PKH, P2WPKH	A key hash alone is not exposure; classification requires local spend/reuse evidence.
Script context required	P2SH, P2WSH	Redeem/witness script review; signer sets, thresholds, timelocks preserved in any route.
P2MR-shaped, pre-activation	OP_2 + 32-byte program	Active-rules and BIP-revision check; never described as mainnet-ready.

7 Phase 2 — optional, proposed only after Phase 1 acceptance

If Phase 1 delivers, I will propose a separately gated Phase 2 in the USD 150,000–300,000 range: a production-grade wallet/custodian migration-readiness analyzer built on the Phase 1 classifier; descriptor and PSBT preflight tooling (regtest/signet only); a measured FIPS 204 (ML-DSA) / FIPS 205 (SLH-DSA) witness-cost benchmark under current consensus limits; and externally reviewed synthesis into a Quantum Migration Reference Architecture. Nothing about Phase 2 is part of this ask, and Phase 1 is complete and useful on its own — the corpus, matrix, and report stand whether or not Phase 2 ever happens.

8 What exists versus what is promised

Exists now, clickable: the public pinned verifier (9/9 official vectors byte-for-byte, including independent recomputation of the misuse vector's documented secp256k1 tweak; 8/8 boundary tests; a 24-case adversarial mutation corpus in which every case carries its specification citation — the beginning of the M1 deliverable, built unpaid), this proposal, and prior shipped open-source repositories. **Promised, gated:** everything in §1. No team, users, partners, letters of intent, or co-funding are claimed. References: Galaxy announcement 2026-07-21; bitcoin/bips BIP 360 pinned at ed4ffcb6..., BIP 340, BIP 174, BIP 3; NIST FIPS 204/205 (Aug 2024); Bitcoin Optech quantum-resistance index; Delving Bitcoin BIP 360 threads; the public bitcoinjs-lib P2MR pull request and independent Bitcoin Core regtest work (pinned per-target at M2).